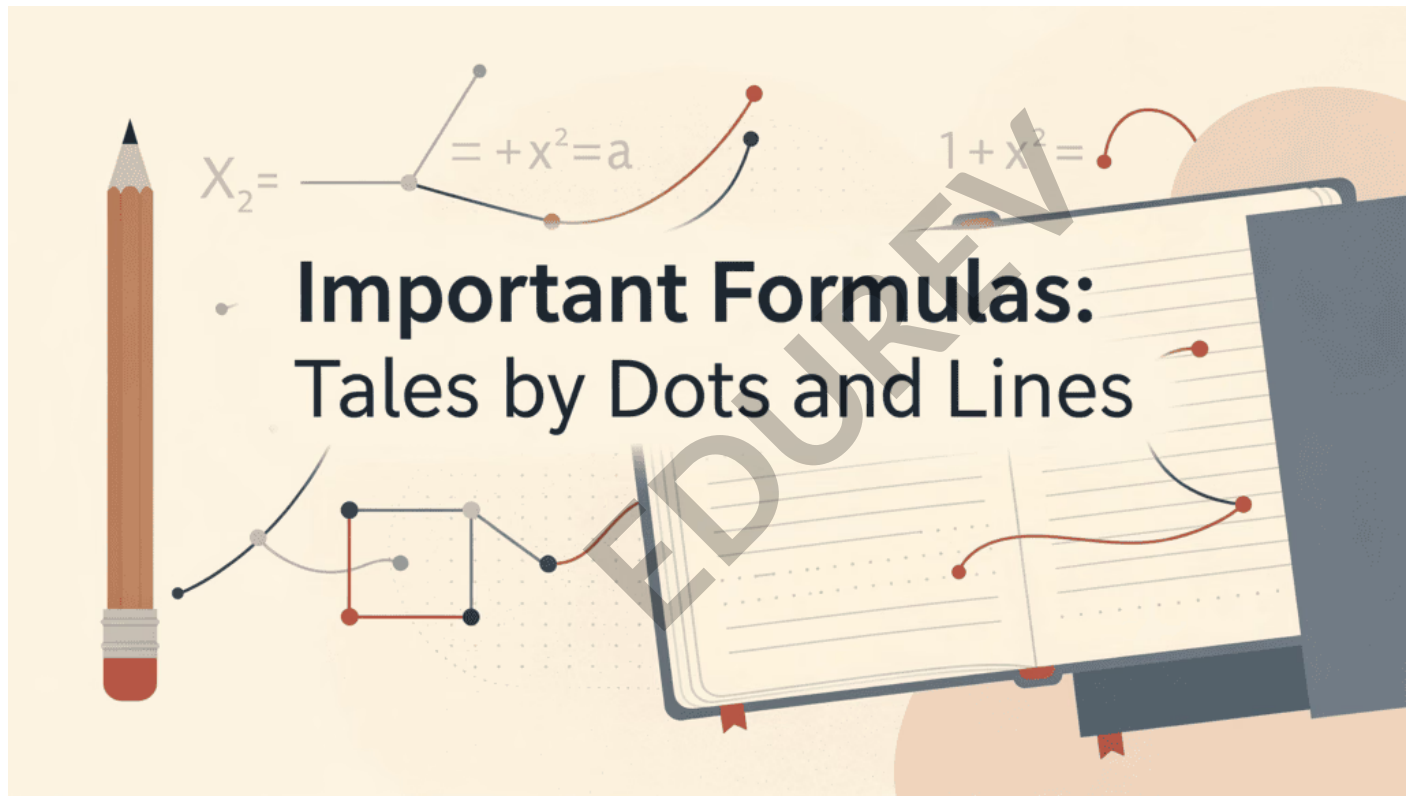




Important Formulas: Tales by Dots and Lines



Formula for Mean (Arithmetic Mean)

Arithmetic Mean Formula

$$\text{Arithmetic Mean} = \frac{\sum_1^n x_i}{n}$$

$$\text{Arithmetic Mean} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

$$\text{Arithmetic Mean} = \frac{\sum_1^n f_i x_i}{n}$$

The mean, or arithmetic mean, is obtained by dividing the sum of all the values by the number of values.

Formula:

$$\text{Mean} = \frac{\text{Sum of all values}}{\text{Number of values}}$$

If the n values are $x_1, x_2, x_3, \dots, x_n$, then:

$$\text{Mean} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

Here, $x_1, x_2, x_3, \dots, x_n$ are the n values in the data.

How do we calculate the mean when frequencies are given?

Ans: When frequencies are given, multiply each value by its frequency, add all these products, and divide the result by the total frequency.

Formula:

$$\text{Mean} = \frac{x_1f_1 + x_2f_2 + x_3f_3 + \dots + x_nf_n}{f_1 + f_2 + f_3 + \dots + f_n}$$

Here, $x_1, x_2, \dots, x_n$ are the values and $f_1, f_2, \dots, f_n$ are their respective frequencies.

Median

$$\text{If 'n' is odd: Median} = \left(\frac{n+1}{2}\right)^{\text{th}} \text{ term}$$

$$\text{If 'n' is even: Median} = \frac{\left(\frac{n}{2}\right)^{\text{th}} \text{ term} + \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ term}}{2}$$

The median is the middle value when data is arranged in order.

First, arrange the data in ascending or descending order. Then, use the number of values to identify the middle position.

For an odd number of values, n is odd:

$$\text{Median} = \text{value at position } \frac{n+1}{2}$$

For an even number of values, n is even:

The median is the average of the values at positions $\frac{n}{2}$ and $(\frac{n}{2} + 1)$.

$$\text{Median} = \frac{\text{value at position } \frac{n}{2} + \text{value at position } (\frac{n}{2} + 1)}{2}$$

Example: For the data 3, 5, 7, 9, 11, there are 5 values. The median is the value at position $\frac{5+1}{2} = 3$, which is 7.

How does the mean change when a fixed number is added to every value?

Ans: The new mean is the old mean plus the fixed number.

Formula: If the original mean is a , and k is added to every value, then:

$$\text{New mean} = a + k$$

When k is added to all the values:

$$\begin{aligned}\text{New mean} &= \frac{(x_1 + k) + (x_2 + k) + \dots + (x_n + k)}{n} \\ &= \frac{x_1 + x_2 + \dots + x_n + nk}{n} \\ &= \frac{x_1 + x_2 + \dots + x_n}{n} + k \\ &= a + k\end{aligned}$$

How does the mean change when a fixed number is subtracted from every value?

Ans: The new mean is the old mean minus the fixed number.

Formula: If the original mean is a , and k is subtracted from every value, then:

$$\text{New mean} = a - k$$

How does the mean change when every value is multiplied by a fixed number?

Ans: The new mean is the old mean multiplied by the fixed number.

Formula: If the original mean is a , and every value is multiplied by k , then:

$$\text{New mean} = k \times a$$

When every value is multiplied by k :

$$\begin{aligned}\text{New mean} &= \frac{kx_1 + kx_2 + \dots + kx_n}{n} \\ &= \frac{k(x_1 + x_2 + \dots + x_n)}{n} \\ &= k \times \frac{x_1 + x_2 + \dots + x_n}{n} \\ &= k \times a\end{aligned}$$

How does the mean change when every value is divided by a fixed number?

Ans: The new mean is the old mean divided by the fixed number.

Formula: If the original mean is a , and every value is divided by k , where $k \neq 0$, then:

$$\text{New mean} = \frac{a}{k}$$

How do we find an unknown value when the mean is given?

Ans: First find the total sum by multiplying the mean by the number of values. Then subtract the sum of the known values from the total sum.

Formula: If the mean is m , the number of values is n , and one value w is unknown, then:

$$w = (m \times n) - \text{sum of all other known values}$$

Example: The weights are 42, 40, 39, 33, 48, 38, 42, 35, 32, w , and their mean is 39.2.

The number of values is 10. Therefore:

$$\begin{aligned}w &= (39.2 \times 10) - (42 + 40 + 39 + 33 + 48 + 38 + 42 + 35 + 32) \\ &= 392 - 349 \\ &= 43 \text{ kg}\end{aligned}$$

How do we find the total sum when the mean is given?

Ans: Multiply the mean by the number of values.

Formula:

$$\text{Total sum} = \text{Mean} \times n$$

Example: If the mean of 15 values is 134, then:

$$\text{Total sum} = 134 \times 15 = 2,010$$

Properties of Mean

Balancing property: The sum of the distances of all values to the left of the mean is equal to the sum of the distances of all values to the right of the mean.

Key points:

Including a value greater than the mean increases the mean.

Including a value less than the mean decreases the mean.

Including a value equal to the mean keeps the mean unchanged.

Removing a value greater than the mean decreases the mean.

Removing a value less than the mean increases the mean.

Spreadsheet Formulas

Sum formula: =SUM(Start:End)

Example: =SUM(B3:G3) calculates the total of the cells from B3 to G3.

Average formula: =AVERAGE(Start:End)

Example: =AVERAGE(B7:D7) calculates the average of the cells from B7 to D7.

Cell range notation:

Row range: B3:G3 represents all the cells from B3 to G3 in the same row.

Column range: D2:D23 represents all the cells from D2 to D23 in the same column.

